A CONSTRUCTIBLE COHOMOLOGY SHEAF OF A COMPLEX IS HIT BY A FINITE SUM OF EXTENSIONS BY ZERO OF THE CONSTANT SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the projection of the kernel onto the cohomology sheaf, the surjection from extension sheaves over affine objects, and the finiteness step from constructibility.

Editorial note. The label lemma-one-constructible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Surjection onto a module from a direct sum of extensions by zero. Modules on Sites, the lemma representing a sheaf of modules as a quotient of a direct sum of j_{U!} of the structure sheaf over affine objects. Supplies the surjection onto the kernel of K^b -> K^{b+1} used at the start.
- Constructible sheaves are finitely generated over the colimit. Same chapter, the lemma stating that a constructible sheaf is the union of the images of maps from finitely many extension sheaves. Supplies that the image of a finite sub-sum already equals H^b(K).

Source reference. Original reference 095P: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11903–11903. Prerequisite pointer invoked at proof line(s) 13637; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 093B: sources/stacks-chow/tags.txt, source lines 11509–11509. Prerequisite pointer invoked at proof line(s) 13630; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let Λ be a Noetherian ring. Let $K \in D(X_{étale}, \Lambda)$ and $b \in \mathbf{Z}$ such that $H^b(K)$ is constructible. Then there exist a sheaf $\mathcal{F}$ which is a finite direct sum of $j_{U!}\underline{\Lambda}$ with $U \in \mathrm{Ob}(X_{étale})$ affine and a map $\mathcal{F}[-b] \to K$ in $D(X_{étale}, \Lambda)$ inducing a surjection $\mathcal{F} \to H^b(K)$.*

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Proof. Represent K by a complex $\mathcal{K}^\bullet$ of sheaves of Λ-modules. Consider the surjection

$$\mathrm{Ker}(\mathcal{K}^b \to \mathcal{K}^{b+1}) \longrightarrow H^b(K)$$

By Modules on Sites, Lemma 093B we may choose a surjection $\bigoplus_{i\in I} j_{U_i!}\underline{\Lambda} \to \mathrm{Ker}(\mathcal{K}^b \to \mathcal{K}^{b+1})$ with U_i affine. For $I' \subset I$ finite, denote $\mathcal{H}_{I'} \subset H^b(K)$ the image of $\bigoplus_{i\in I'} j_{U_i!}\underline{\Lambda}$. By Lemma 095P we see that $\mathcal{H}_{I'} = H^b(K)$ for some $I' \subset I$ finite. The lemma follows taking $\mathcal{F} = \bigoplus_{i\in I'} j_{U_i!}\underline{\Lambda}$. □